

Peter Xu

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<https://xretep.github.io/pdfs/>

EDUCATION

2017 - 2023 PhD in Mathematics at **McGill University**

2012 - 2016 BA in Mathematics at the **University of Chicago** (minor: Human Rights)

EMPLOYMENT

2023 - present Postdoc (with various nominal title changes) at **UCLA**

COMPLETE LIST OF ARTICLES AND PREPRINTS

Organized thematically.

Eisenstein cocycles for Drinfeld modular varieties.

Joint with Romyar Sharifi, preprint. Constructs cocycles valued in higher analogues of Beilinson–Kato elements in the motivic cohomology of Drinfeld modular varieties. Proves Hecke equivariance properties for these cocycles; in particular, in certain dimensions, proving the Eisenstein property of Sharifi’s map ϖ_N relating the Eisenstein geometry of $\mathrm{GL}_n(A)$ and the motivic cohomology of a rank- $(n - 1)$ Drinfeld modular variety over A , for A any function field. In the setting $n = 2$, shows additional expected properties (integrality, “full” parabolicity, U_p^* -invariance) for this map in full analogy to the classical cyclotomic setting of Sharifi’s conjectures. Available at

<https://xretep.github.io/pdfs/eisdrinfeld.pdf>.

Symbols for toric Eisenstein cocycles and arithmetic applications

Submitted. Obtains explicit expressions for Eisenstein cocycles for arithmetic group actions on self-products of $\mathbf{G}_m$, by constructing explicit symbols in motivic cohomology complexes. Improves existing Hecke and integrality results for Sharifi’s map ϖ_N relating the Eisenstein geometry of $\mathrm{GL}_2(\mathbb{Q})$ and the motivic cohomology of cyclotomic fields. Establishes a duality relationship of the de Rham regulator to smoothed Shintani generating functions, and thereby obtains a new Shintani-style proof of the 2-adic congruences of Deligne–Ribet. Available at <https://xretep.github.io/pdfs/toric.pdf>.

Geometric model for the integral cohomology of toric hypersurface arrangements

Preprint. Gives a self-contained proof of a geometric presentation for the integral cohomology ring of an arbitrary toric hypersurface complement, in terms of elementary conical chamber data. Describes explicit Borel–Moore cycle and log-differential form representatives; the latter gives a new, independent proof of formality over $\mathbb{Q}$ of such arrangements. Our description refines and simplifies a line of previous results in the toric arrangement literature, notably culminating in the work of Callegaro et al. (who give a more complicated presentation which we show is equivalent to ours). Available at

<https://xretep.github.io/pdfs/cones.pdf>.

A note on polyhedral cones and toric polylogarithms

Submitted. Gives an explicit combinatorial-geometric presentation of the trace-fixed part of the Gersten complex for the function field of an algebraic torus over $\mathbb{Q}$.

Available at <https://xretep.github.io/pdfs/spheres.pdf>.

A note on cubical Bloch–Levine cycle complexes

Submitted. Checks that Levine’s argument that the cubical Bloch complex computes the same cohomology as its simplicial analogue also holds over semi-local PIDs, and in particular that the sheaf of cubical complexes computes motivic cohomology for smooth schemes. Available at

<https://xretep.github.io/pdfs/cubical.pdf>.

Localization and elliptic motivic relations

Submitted. Derives from a general motivic localization formalism a large generalization of classical Suslin/Weil–style reciprocity laws, including in the setting of motivic homotopy theory. Uses this to construct explicit modular symbols valued in integral-coefficient motivic cohomology of some arithmetic families of group schemes with arbitrary level structures, including symbols valued in Beilinson–Kato elements and Eisenstein symbols, as well as some new candidates for Sharifi’s map in the group homology-to- K -theory direction.

Available at <https://xretep.github.io/pdfs/suslin.pdf>.

Elliptic polylogarithms and modular forms

Preprint. This note describes the construction and some applications of an equivariant construction of a holomorphic realization of elliptic polylogarithms, parameterized explicitly by Steinberg symbols. The resulting families of modular forms have connections to many arithmetic and automorphic phenomena, including toric modular forms and rigid analytic cocycles.

Available at <https://xretep.github.io/pdfs/fay.pdf>.

Presenting restricted Steinberg modules of general linear groups

Preprint. Defines “restricted” versions of Steinberg modules for general linear groups coming from local restrictions on cusps, and proves analogues of Borel–Serre’s results relating their coinvariants with the geometry of partial compactifications of associated locally symmetric spaces. Gives generators and presentations for particularly nice cases of these restricted Steinberg modules.

Available at <https://xretep.github.io/pdfs/restricted.pdf>.

Eisenstein class of a torus bundle and log-rigid classes for $\mathrm{SL}_n(\mathbb{Z})$

Joint with Martí Roset Julià, submitted. Uses purely topological methods (part of the “topological polylogarithm” class, to be precise) to produce a $(n - 1)$ -cocycle for $\mathrm{SL}_n(\mathbb{Z})$ valued in differential forms on Drinfeld’s p -adic upper half-space. Conjectures a relationship between evaluations of these classes and logarithms of Gross–Stark units, and proves this in some cases using the work of Dasgupta–Kakde.

Available at <https://xretep.github.io/pdfs/RX1.pdf>.

Rigid analytic Eisenstein cocycles for $\mathrm{SL}_n(\mathbb{Z}[1/p])$.

Joint with Martí Roset Julià, preprint. Uses symbols in a cohomological complex for a smooth torus to produce explicit $(n - 1)$ -cocycles for $SL_n(\mathbb{Z}[1/p])$ valued in invertible analytic functions on Drinfeld's p -adic upper half-space, generalizing existing results on rigid analytic cocycles from the case $n = 2$, in the “Eisenstein” setting. Relates the evaluations to periods of Eisenstein series. Proves that the special moduli of these cocycles yield Brumer–Stark units for totally real fields in which p is inert, using the integral Gross–Stark conjecture; provides evidence the general case relates to boundary degenerations of these same geometric objects. Available at <https://xretep.github.io/pdfs/RX2.pdf>.

Classical periods of Eisenstein series and Bernoulli polynomials in the equivariant cohomology of a torus

Published in *Research in Number Theory*. Uses explicit elements in cohomology complexes of 2-tori to obtain a conceptual proof of a classical *unsmoothed* formula of Siegel for the periods of classical weight-2 holomorphic Eisenstein series. Available at <https://xretep.github.io/pdfs/classical.pdf>.

Explicit formula for the (GL_2, GL_2) theta lift via Bruhat decomposition

Submitted. Uses explicit elements in cohomology complexes of the universal elliptic curve to obtain explicit unsmoothed formulas for the Eisenstein theta lift for $(GL_2/\mathbb{Q}, GL_2/\mathbb{Q})$.

Available at <https://xretep.github.io/pdfs/bruhat.pdf>.

RESEARCH OUTREACH

Workshop on rigid cocycles

Oberwolfach, March 2027

Upcoming Oberwolfach seminar, whose material in part covers my work (also joint with Julià) on rigid Eisenstein cocycles and their role in Hilbert's 12th problem.

Models for polylogarithms over cyclotomic fields

ICTS Bengaluru, January 2027

Upcoming talk at the ICTS program *Fifty Years of Eisenstein Ideal*, on my work on the relation of combinatorial models of hypersurface arrangements to recent developments in Goncharov's program.

Geometric arrangements and class field theory

CMS Montreal, December 2026

Upcoming talk on my work, in relation to the role of Steinberg modules in recent developments in Goncharov's program and adjacent phenomena.

Combinatorial Eisenstein cocycles

2024 Bielefeld Workshop

Talk on the construction of explicit Eisenstein cocycles in the toric/elliptic cases (preprints mentioned above).

Higher Kato elements and equivariant complexes

2022 Quebec-Maine Number Theory

Talk on a part of my thesis work in a concrete setting.

Deformations of formal groups

2018 Montreal-Toronto Number Theory

Expository talk on the Gross-Hopkins construction of deformation spaces of formal groups.

Various seminar talks

Columbia, Caltech, UCSB, UCLA, etc.

Talks on various aspects of my research, over the span of a few years.

AWARDS AND SCHOLARSHIPS

McGill Graduate Excellence Award 2017 – 2023

Kenneth Eade Fellow 2018 – 2019

McGill Chair Graduate Award 2021 – 2022

TEACHING AND MENTORING

Lectureships

McGill University:

- MATH 141 Calculus II, Summer 2018 and Fall 2022

University of California, Los Angeles:

- MATH 131a Analysis, Fall 2023, Winter/Spring/Fall 2024, Winter/Fall 2025
- MATH 184 Enumerative Combinatorics, Spring 2024
- MATH 61 Introduction to Discrete Structures, Winter 2025
- MATH 111 Introduction to Number Theory, Spring 2025
- MATH 132 Complex Analysis, Fall 2026 (in progress)

Directed Reading Program

Organized a mentorship program pairing McGill undergraduates interested in mathematics research with graduate students in corresponding fields. Solicited and paired graduate student volunteers with interested undergraduates, organized events, reviewed student expository write-ups, and participated as a mentor (in the first two years). The program has now expanded to be well-funded, with stipends for the graduate mentors, funds for resources for the students, and writing/presentation workshops.

UCLA–CSU Summer Bridge

August 2025. Prepared and taught an intensive one-week course (seven hours per day, with breaks) on representation theory of finite groups for a small group of students preparing to apply for doctorates in mathematics, coming from California State system universities which traditionally do not have strong programs for mathematical research. Wrote syllabus, assigned exercises, and gave personalized attention to each student throughout the long teaching sessions.

LANGUAGES

Fluent English, French, Mandarin Chinese

Conversational Spanish, German